

Incidence of Strabismus Post-Plaque Brachytherapy in Patients With Uveal Melanoma



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- **PURPOSE:** To determine the incidence and type of strabismus in patients with uveal melanoma treated with plaque brachytherapy.
- **DESIGN:** Multicenter, retrospective incidence estimation study.
- **METHODS:** A total of 438 eyes of 438 patients with uveal melanoma treated with plaque brachytherapy between October 2011 and May 2021. Intervention was Iodine 125, and Palladium 103 plaque brachytherapy. The variables reviewed included incidence of nonresolving strabismus post-plaque brachytherapy, type of strabismus developed, extraocular muscles operated, and modality of treatment received.
- **RESULTS:** A total of 438 patients underwent plaque brachytherapy treatment for uveal melanoma. Eleven patients developed strabismus post-plaque brachytherapy (2.5%, $n = 11/438$). Of these patients, 5 (1.1%, $n = 5/438$) developed strabismus immediately postoperation. Specifically, 2 patients (0.5%, $n = 2/438$) developed strabismus immediately postoperation due to slipped muscles, 2 patients (0.5%, $n = 2/438$) due to decompensated phorias, and 1 patient (0.5%, $n = 1/438$) due to a fibrotic muscle. Six patients (1.4%, $n = 6/438$) developed late-onset sensory strabismus. A total of 355 patients (81.1%, $n = 355/438$) had their extraocular muscles disinserted during surgery, with the lateral rectus being the most common, accounting for 45.4% ($n = 161/355$), followed by the superior rectus at 26.8% ($n = 95/355$). Strabismus surgery was the most common treatment modality, comprising 72.7% ($n = 8/11$) of patients.
- **CONCLUSIONS:** The incidence of strabismus after plaque brachytherapy treatment for uveal melanoma was

low and primarily classified as late-onset sensory strabismus. Previous studies may underestimate the long-term incidence of strabismus after plaque brachytherapy by focusing primarily on strabismus present immediately postoperatively. (Am J Ophthalmol 2024;262: 161–169. © 2024 Elsevier Inc. All rights reserved.)

PLAQUE BRACHYTHERAPY IS THE MOST WIDELY USED therapy for uveal melanoma.^{1–16} Briefly, plaque brachytherapy is a radiation therapy technique involving the surgical placement of a radioactive implant (Iodine-125/Palladium-103/Ruthenium-106) overlying the site of the intraocular tumor.^{1–11} Often, to optimize radiation to the melanoma or to ensure adequate visualization, extraocular muscles must be disinserted (Figure 1).^{17–19}

Strabismus and diplopia have been reported as rare complications after plaque radiation therapy.^{19–22} Nonresolving strabismus noted immediately after plaque surgery has been reported in 1.9% to 2.8% of cases.^{19–22} Possible causes include direct mechanical injury to extraocular muscles during plaque placement, suture release around the time of surgery (slipped muscle), and ionizing radiation damage to surrounding ocular tissue.²² To date, few studies have reported on long-term results, including late-onset strabismus, and we could not identify any studies that included late-onset strabismus and the categorization of the type of strabismus.^{19–22} Because of vision loss that can occur after treatment, late-onset strabismus, such as sensory strabismus, is an important complication that patients need to be informed of. The purpose of this study was to determine the incidence and type of strabismus in patients post-plaque brachytherapy, including late-development strabismus. Secondary aims include identification of the most common treatment modality provided to patients who developed strabismus postoperation.

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METHODS

- **STUDY DESIGN:** The study was a retrospective incidence estimation study derived from two centers in Edmonton and Calgary, Alberta, Canada.